

Mohammed Abrar

Full-Stack Developer · Backend Engineer

📍 Addis Ababa, Ethiopia 📞 0953 442 423 ✉ mame.534424@gmail.com 📄 github.com/mame534424
🌐 www.linkedin.com/in/mohammed-abrar-ba0291312

PROFILE

Computer Engineering student at Addis Ababa University and backend-focused Full-Stack Developer passionate about building scalable, production-grade systems. Experienced in secure web and mobile applications using React, React Native, Spring Boot, and Express. Built real-world projects including a real-time chat platform, a fuel queue management system, and an AI-powered transformer monitoring system. Strong interest in backend architecture, API design, AI integration, and system engineering.

EXPERIENCE

Full-Stack Engineer · Lumen Labs — Addis Ababa

Oct 2025 – Present

- Developed and deployed 2 professional client websites with responsive UI, SEO optimization, and high performance.
- Implemented secure backend features including RBAC, JWT authentication workflows, and protected API routes on large-scale web applications.
- Optimized database queries and backend workflows, significantly improving response time and loading performance.
- Designed backend API architecture and authored technical endpoint documentation for frontend and mobile integration teams.
- Collaborated on security hardening, validation logic, and scalable system design using React, Express, Spring Boot, REST APIs, PostgreSQL, and MongoDB.

PROJECTS

E-Fuel — Fuel Availability Tracking & Queue Management System

Tech Stack: Express.js · Next.js · React Native · PostgreSQL

A full-stack platform addressing Ethiopia's fuel shortages and long queue problem, serving three user groups: government administrators, fuel stations, and the public.

- Built real-time fuel availability tracking with queue monitoring and station map integration.
- Developed a fuel reservation/booking system with user notifications and real-time queue alerts.
- Created government admin dashboard for fuel distribution monitoring and transparency across the supply chain.
- Built station manager dashboards for managing fuel availability and customer queues.
- Developed mobile app (React Native) for users to locate stations, book fuel, and receive live notifications.

Role: Sole full-stack developer — database design, backend APIs, mobile app, web dashboard, auth, and deployment.

Buna ChatApp — Real-Time Messaging Platform

Tech Stack: Spring Boot · MySQL · React · WebSockets · Gemini LLM · AWS S3

A modern real-time messaging platform inspired by Telegram with an Ethiopian-themed UX, supporting personal/group chats, multimedia sharing, and an AI-powered assistant.

- Implemented WebSocket/STOMP-based real-time messaging for one-to-one and group chat.

- Built audio, image, video, and document sharing with AWS S3-compatible cloud storage.
- Integrated an AI chatbot powered by Gemini directly into the chat experience.
- Designed scalable backend APIs with secure JWT authentication and role-based access control.

Role: Core messaging system, WebSocket communication, file sharing, chatbot integration, backend APIs, and frontend-backend integration.

Smart Transformer Health Monitoring System

Tech Stack: Next.js · FastAPI · Python · Random Forest · LangChain · Vector DB · RAG

An AI-powered platform that predicts transformer health conditions and delivers intelligent maintenance recommendations using machine learning and RAG-based engineering assistance.

- Trained Random Forest ML model for transformer fault analysis and health status prediction.
- Built a RAG-powered chatbot using LangChain + Vector DB, ingesting IEEE research papers to answer engineering questions.
- Developed FastAPI backend services and an interactive Next.js analytics dashboard.
- End-to-end system design: data preprocessing, model training, RAG pipeline, API, and frontend.

Role: Independently designed and built the entire system from data preprocessing through deployment.

SKILLS

Languages: Java · TypeScript · JavaScript · Python · SQL

Frontend: React · Next.js · React Native · Tailwind CSS · HTML/CSS

Backend: Spring Boot · Node.js · Express.js · FastAPI · REST APIs · WebSockets

Databases: PostgreSQL · MySQL · MongoDB

AI & ML: Random Forest · Scikit-learn · LangChain · RAG Architecture · Gemini LLM · Vector DB

Auth & Security: JWT Authentication · RBAC · OAuth2 · Protected Routes

DevOps & Tools: Docker · Git · GitHub · Postman · AWS S3 · Prometheus · Grafana

Testing: Playwright · Selenium · End-to-End Testing

EDUCATION

B.Sc. Electrical and Computer Engineering

Jan 2023 – Jan 2028

Addis Ababa University, Ethiopia

Everest Youth Academy

Completed 2022

Sheger City · High scorer in Sheger City

Top 10 scorer in Oromia on the Ethiopian University Entrance Examination (EUEE)